



Humans not only take debts to manage necessities. A country may also take debt to manage its economy. For example, infrastructure spending is one costly ingredient required for a country's citizens to lead comfortable lives. The World Bank is the organization that provides debt to countries.

In this project, you are going to analyze international debt data collected by The World Bank. The dataset contains information about the amount of debt (in USD) owed by developing countries across several categories. You are going to find the answers to the following questions:

- What is the number of distinct countries present in the database?
- What country has the highest amount of debt?
- What country has the lowest amount of repayments?

Below is a description of the table you will be working with:

`international_debt` table

Column	Definition	Data Type
country_name	Name of the country	varchar
country_code	Code representing the country	varchar
indicator_name	Description of the debt indicator	varchar
indicator_code	Code representing the debt indicator	varchar
debt	Value of the debt indicator for the given country (in current US dollars)	float

You will execute SQL queries to answer three questions, as listed in the instructions.

```
SELECT *
FROM international_debt
```

...	↑↓	country_name	...	↑↓	countr...	...	↑↓	indicator_name	...	↑↓	indicator_...	...	↑↓	d
1769		Russian Federation			RUS			Disbursements on external debt, long-term (DIS, ...			DT.DIS.DLXF.CD			▲
1770		Russian Federation			RUS			Interest payments on external debt, long-term (l...			DT.INT.DLXF.CD			
1771		Russian Federation			RUS			Interest payments on external debt, private non...			DT.INT.DPNG.CD			
1772		Russian Federation			RUS			PPG, bilateral (AMT, current US\$)			DT.AMT.BLAT.CD			
1773		Russian Federation			RUS			PPG, bilateral (INT, current US\$)			DT.INT.BLAT.CD			
1774		Russian Federation			RUS			PPG, bonds (AMT, current US\$)			DT.AMT.PBND.CD			
1775		Russian Federation			RUS			PPG, bonds (INT, current US\$)			DT.INT.PBND.CD			
1776		Russian Federation			RUS			PPG, commercial banks (AMT, current US\$)			DT.AMT.PCBK.CD			
1777		Russian Federation			RUS			PPG, commercial banks (INT, current US\$)			DT.INT.PCBK.CD			
1778		Russian Federation			RUS			PPG, multilateral (AMT, current US\$)			DT.AMT.MLAT.CD			
1779		Russian Federation			RUS			PPG, multilateral (DIS, current US\$)			DT.DIS.MLAT.CD			
1780		Russian Federation			RUS			PPG, multilateral (INT, current US\$)			DT.INT.MLAT.CD			
1781		Russian Federation			RUS			PPG, official creditors (AMT, current US\$)			DT.AMT.OFFT.CD			
1782		Russian Federation			RUS			PPG, official creditors (DIS, current US\$)			DT.DIS.OFFT.CD			
1783		Russian Federation			RUS			PPG, official creditors (INT, current US\$)			DT.INT.OFFT.CD			
1784		Russian Federation			RUS			PPG, other private creditors (AMT, current US\$)			DT.AMT.PROP.CD			▼

```
-- count distinct countries

-- Define a query named 'num_distinct_countries'
WITH num_distinct_countries AS (

    -- Select distinct countries, count them, and provide output as total_distinct_countries, in one row and column
    SELECT
        COUNT(DISTINCT country_name) AS total_distinct_countries
    FROM
        international_debt
)

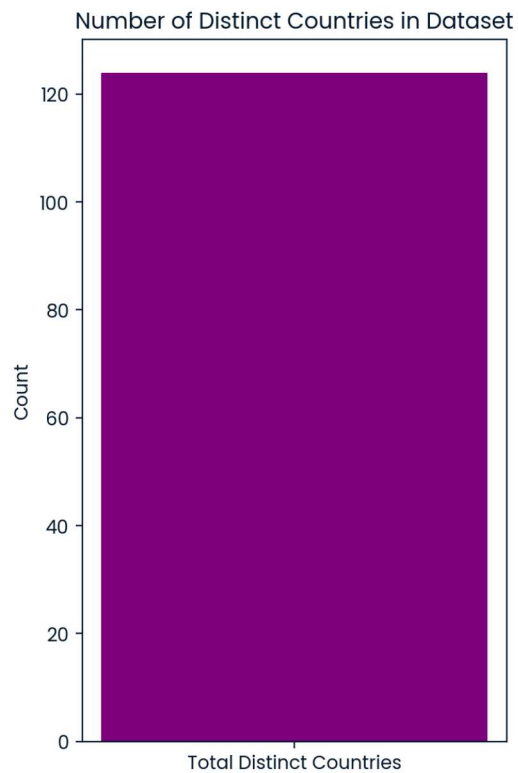
-- Select the result from the query
SELECT *
FROM num_distinct_countries;
```

index	...	↑↓	total_distinct_countries
			0

```
import pandas as pd
import matplotlib.pyplot as plt

# Step 1: Count the number of distinct countries
num_distinct_countries_df = original_df['country_name'].nunique()

# Step 2: Visualize using a simple bar chart
plt.figure(figsize=(4, 6))
plt.bar(['Total Distinct Countries'], [num_distinct_countries_df], color='purple')
plt.title('Number of Distinct Countries in Dataset')
plt.ylabel('Count')
plt.tight_layout()
plt.show()
```



```
-- To calculate highest total disbursements for each country where indicator_code = DT.DIS.DLXF.CD, and include the country's
repayment amount where indicator_code = DT.AMT.DLXF.CD
```

```
-- CODE, using a CTE...
-- Define a query named 'highest_disbursements'
```

```
WITH highest_disbursements AS (
  SELECT country_name,
    SUM(CASE
      WHEN indicator_code = 'DT.DIS.DLXF.CD' THEN debt
      ELSE 0
    END) AS total_disbursements,
    SUM(CASE
      WHEN indicator_code = 'DT.AMT.DLXF.CD' THEN debt
      ELSE 0
    END) AS total_repayments
  FROM international_debt
  GROUP BY country_name
)
SELECT *
FROM highest_disbursements
ORDER BY total_disbursements DESC
LIMIT 20;
```

index	...	↑↓	country_name	...	↑↓	total_disbursements	...	↑↓	total_repayments
0			Least developed countries: UN classification			40160766261.6			
1			IDA only			34531188113.2			
2			South Asia			29306216064.7			
3			Cameroon			18186662060.4			
4			China			15692563746.1			
5			India			11005547326.2			
6			Angola			10924018093.1			
7			Egypt, Arab Rep.			9552207423.5			
8			Bangladesh			9050557611.9			
9			Vietnam			6494121653.2			
10			Pakistan			5301166769.9			
11			Kenya			4266494939.7			
12			Indonesia			4177287375.8			
13			Brazil			4092388651.4			
14			Ethiopia			2681493007.3			
15			Sri Lanka			2547120639.9			

Rows: 20

Expand

```

import matplotlib.ticker as ticker

plt.figure(figsize=(14, 7))
bar_width = 0.4
x = range(len(highest_disbursements_df))

# Plot disbursements
plt.bar(x, highest_disbursements_df['total_disbursements'],
        width=bar_width, label='Disbursements', color='orange')

# Plot repayments next to disbursements
plt.bar(
    [i + bar_width for i in x],
    highest_disbursements_df['total_repayments'],
    width=bar_width, label='Repayments', color='blue'
)

# X-axis labels
plt.xticks(
    [i + bar_width / 2 for i in x],
    highest_disbursements_df['country_name'],
    rotation=45, ha='right'
)

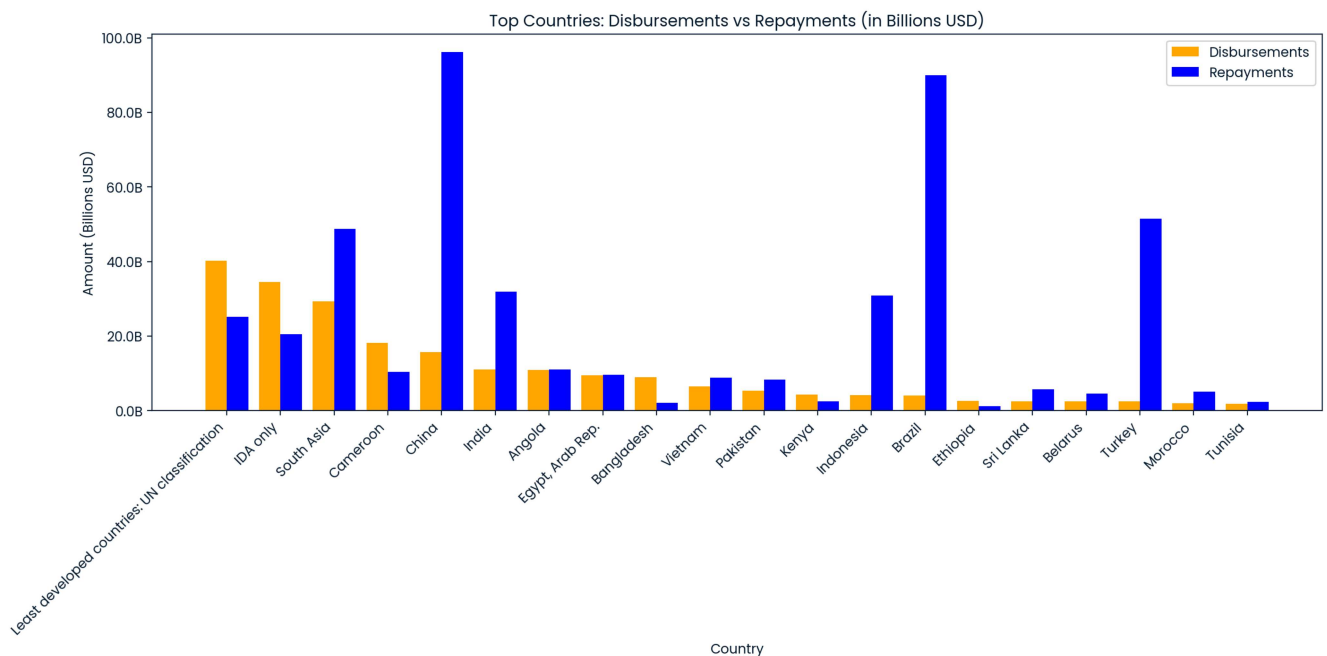
# Y-axis formatting: billions
formatter = ticker.FuncFormatter(lambda x, _: f'{x/1e9:.1f}B')
plt.gca().yaxis.set_major_formatter(formatter)

# Labels and title
plt.ylabel('Amount (Billions USD)')
plt.xlabel('Country')
plt.title('Top Countries: Disbursements vs Repayments (in Billions USD)')
plt.legend()

# Tight layout for better spacing
plt.tight_layout()

# Show plot
plt.show()

```



-- To calculate lowest total disbursements for each country where indicator_code = DT.DIS.DLXF.CD, and include the country's repayment amount where indicator_code = DT.AMT.DLXF.CD

-- CODE, using a CTE...
-- Define a query named 'lowest_disbursements'

```
WITH lowest_disbursements AS (  
  SELECT country_name,  
         SUM(CASE  
           WHEN indicator_code = 'DT.DIS.DLXF.CD' THEN debt  
           ELSE 0  
         END) AS total_disbursements,  
         SUM(CASE  
           WHEN indicator_code = 'DT.AMT.DLXF.CD' THEN debt  
           ELSE 0  
         END) AS total_repayments  
  
  FROM international_debt  
  GROUP BY country_name  
)  
SELECT *  
FROM lowest_disbursements  
ORDER BY total_disbursements ASC  
LIMIT 5; --20 if necessary
```

...	↑↓	country_n...	...	↑↓	total_disburseme...	...	≡↑	total_repay...	...	↑↓
	0	Turkmenistan			0			29132060.9		
	1	Somalia			2456971.6			32985034.3		
	2	Tonga			5529604.8			10369910.6		
	3	Eritrea			7203213.6			31110077.8		
	4	Solomon Islands			8645249.9			30749703.9		

```

plt.figure(figsize=(10, 6))

bar_width = 0.4
x = range(len(lowest_disbursements_df)) # Positions for bars

# Plot disbursements (light blue)
plt.bar(x, lowest_disbursements_df['total_disbursements'],
        width=bar_width, label='Disbursements', color='skyblue')

# Plot repayments (light red), shifted right
plt.bar([i + bar_width for i in x], lowest_disbursements_df['total_repayments'],
        width=bar_width, label='Repayments', color='blue')

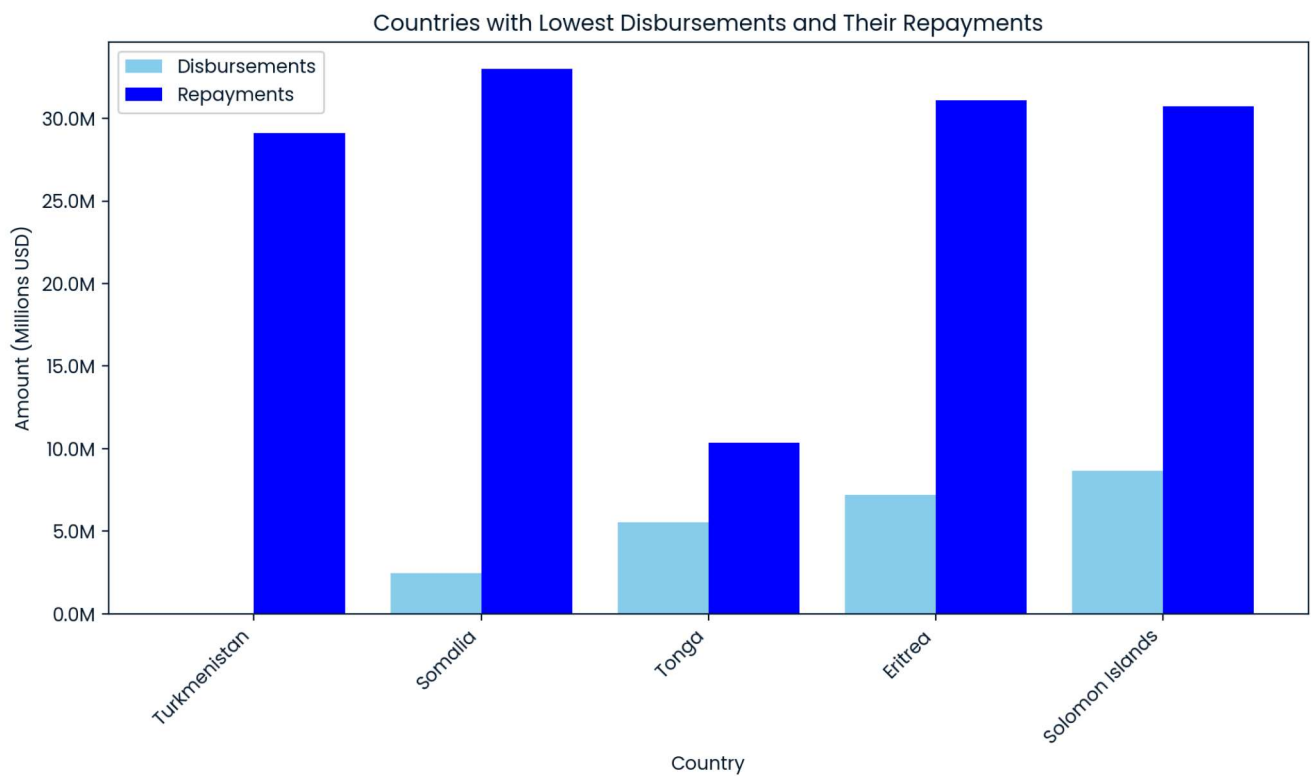
# Add country names below the bars
plt.xticks([i + bar_width / 2 for i in x],
           lowest_disbursements_df['country_name'],
           rotation=45, ha='right')

# Format y-axis in millions
formatter = ticker.FuncFormatter(lambda x, _: f'{x/1e6:.1f}M')
plt.gca().yaxis.set_major_formatter(formatter)

# Add labels and title
plt.xlabel('Country')
plt.ylabel('Amount (Millions USD)')
plt.title('Countries with Lowest Disbursements and Their Repayments')
plt.legend()

plt.tight_layout()
plt.show()

```



```

-- lowest_principal_repayment
-- CODE, using a CTE...
-- Define a query named 'lowest_principal_repayment'

WITH lowest_principal_repayment AS (
  -- Select country_name, indicator_name and debt aliased as lowest_repayment, where indicator_name = DT.AMT.DLXF.CD, in
  ascending order. Limit to 5
  SELECT country_name,

    SUM(CASE
      WHEN indicator_code = 'DT.AMT.DLXF.CD' THEN debt
      ELSE 0
    END) AS total_repayments,

    SUM(CASE
      WHEN indicator_code = 'DT.DIS.DLXF.CD' THEN debt
      ELSE 0
    END) AS total_disbursements

  FROM international_debt
  GROUP BY country_name
)
SELECT *
FROM lowest_principal_repayment
ORDER BY total_repayments ASC
LIMIT 5; -- 20 if necessary

```

...	↑↓	country_name	...	↑↓	total_repay...	...	↑↓	total_disburseme...	...	↑↓
	0	Timor-Leste			825000			21799986.4		
	1	Sao Tome and Principe			2941396.3			10636848.5		
	2	Comoros			4792269.1			13460035.4		
	3	Guinea-Bissau			6783629.1			48265990.2		
	4	Tonga			10369910.6			5529604.8		

Rows: 5


 Expand

```

plt.figure(figsize=(12, 6))

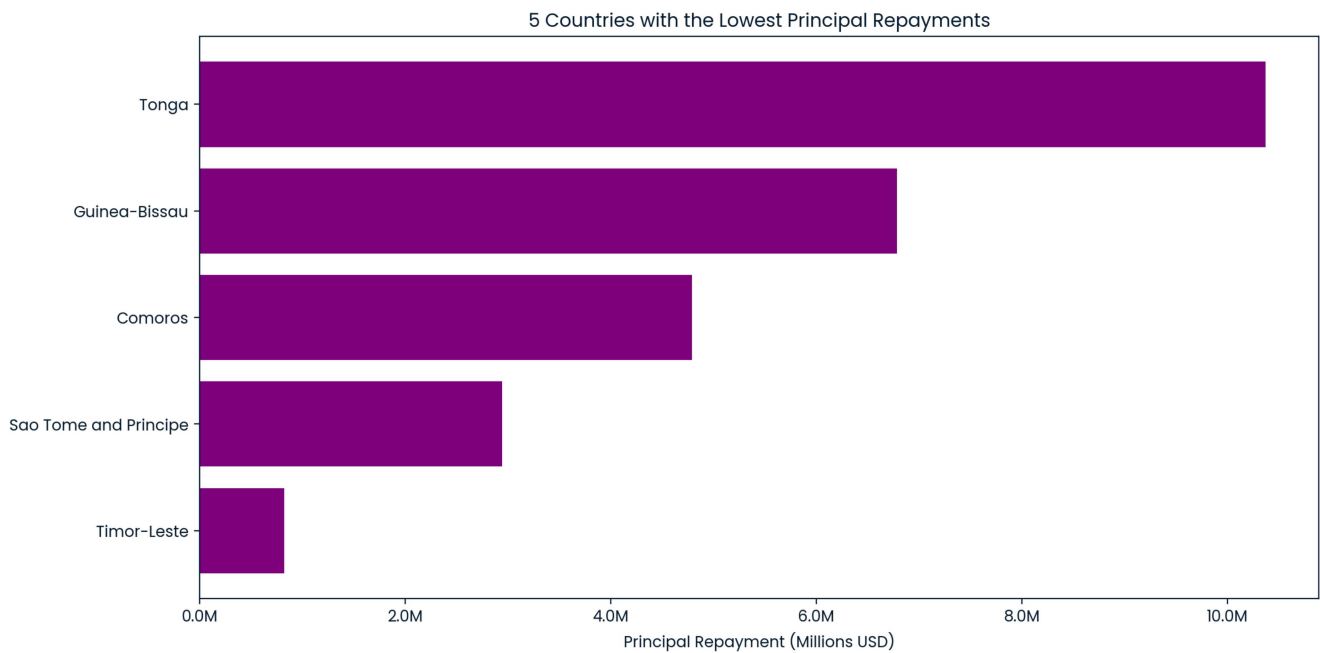
# Create a horizontal bar chart
plt.barh(lowest_principal_repayment['country_name'], lowest_principal_repayment['total_repayments'], color='purple')

# Format y-axis numbers into millions
formatter = ticker.FuncFormatter(lambda x, _: f'{x/1e6:.1f}M')
plt.gca().xaxis.set_major_formatter(formatter)

plt.xlabel('Principal Repayment (Millions USD)')
plt.title('5 Countries with the Lowest Principal Repayments')

plt.tight_layout()
plt.show()

```



```

-- CODE, using a CTE...
-- Define a query named 'highest_repayments'

WITH highest_repayments AS (
    -- Select country_name, indicator_name and debt aliased as highest_repayments, where indicator_name = DT.AMT.DLXF.CD, in
    ascending order. Limit to 20

    SELECT country_name, indicator_name, debt AS highest_repayments
    FROM international_debt
    WHERE indicator_code = 'DT.AMT.DLXF.CD' -- Only principal repayments
    ORDER BY highest_repayments DESC
)
-- Select the result from the query
SELECT *
FROM highest_repayments
Limit 20;

```

...	↑↓	country_name	...	↑↓	indicator_name	...	↑↓	highest_repay...	...	↑↓	
	0	China			Principal repayments on external debt, long-t...			96218620835.7			
	1	Brazil			Principal repayments on external debt, long-t...			90041840304.1			
	2	Russian Federation			Principal repayments on external debt, long-t...			66589761833.5			
	3	Turkey			Principal repayments on external debt, long-t...			51555031005.8			
	4	South Asia			Principal repayments on external debt, long-t...			48756295898.2			
	5	India			Principal repayments on external debt, long-t...			31923507000.8			
	6	Indonesia			Principal repayments on external debt, long-t...			30916112653.8			
	7	Kazakhstan			Principal repayments on external debt, long-t...			27482093686.4			
	8	Mexico			Principal repayments on external debt, long-t...			25218503927			
	9	Least developed countries: UN classification			Principal repayments on external debt, long-t...			25197029299.4			
	10	IDA only			Principal repayments on external debt, long-t...			20483289208			
	11	Romania			Principal repayments on external debt, long-t...			14013783350.4			
	12	Colombia			Principal repayments on external debt, long-t...			11985674438.7			
	13	Angola			Principal repayments on external debt, long-t...			11067045628.1			
	14	Cameroon			Principal repayments on external debt, long-t...			10404814960.2			
	15	Venezuela, RB			Principal repayments on external debt, long-t...			9878659207.2			

Rows: 20


Expand

```
-- to extract countries with the lowest disbursements but highest repayments

-- Get bottom 20 countries by total disbursements
WITH bottom_20_disbursements AS (
  SELECT country_name,
         SUM(CASE WHEN indicator_code = 'DT.DIS.DLXF.CD' THEN debt ELSE 0 END) AS total_disbursements
  FROM international_debt
  GROUP BY country_name
  ORDER BY total_disbursements ASC
  LIMIT 20
),

-- Get top 20 countries by total repayments
top_20_repayments AS (
  SELECT country_name,
         SUM(CASE WHEN indicator_code = 'DT.AMT.DLXF.CD' THEN debt ELSE 0 END) AS total_repayments
  FROM international_debt
  GROUP BY country_name
  ORDER BY total_repayments DESC
  LIMIT 20
)

-- Countries that appear in both lists
SELECT b.country_name,
       b.total_disbursements,
       t.total_repayments
FROM bottom_20_disbursements b
JOIN top_20_repayments t
ON b.country_name = t.country_name;
```

...	↑↓	country_name	...	↑↓	total_disburseme...	...	↑↓	total_repay...	...	↑↓
0		Russian Federation			33043274.4			66589761833.5		

Rows: 1

[Expand](#)

```

total_repayments = highest_repayments_df['highest_repayments'].sum()
highest_repayments_df['repayment_percentage'] = (
    highest_repayments_df['highest_repayments'] / total_repayments * 100
)

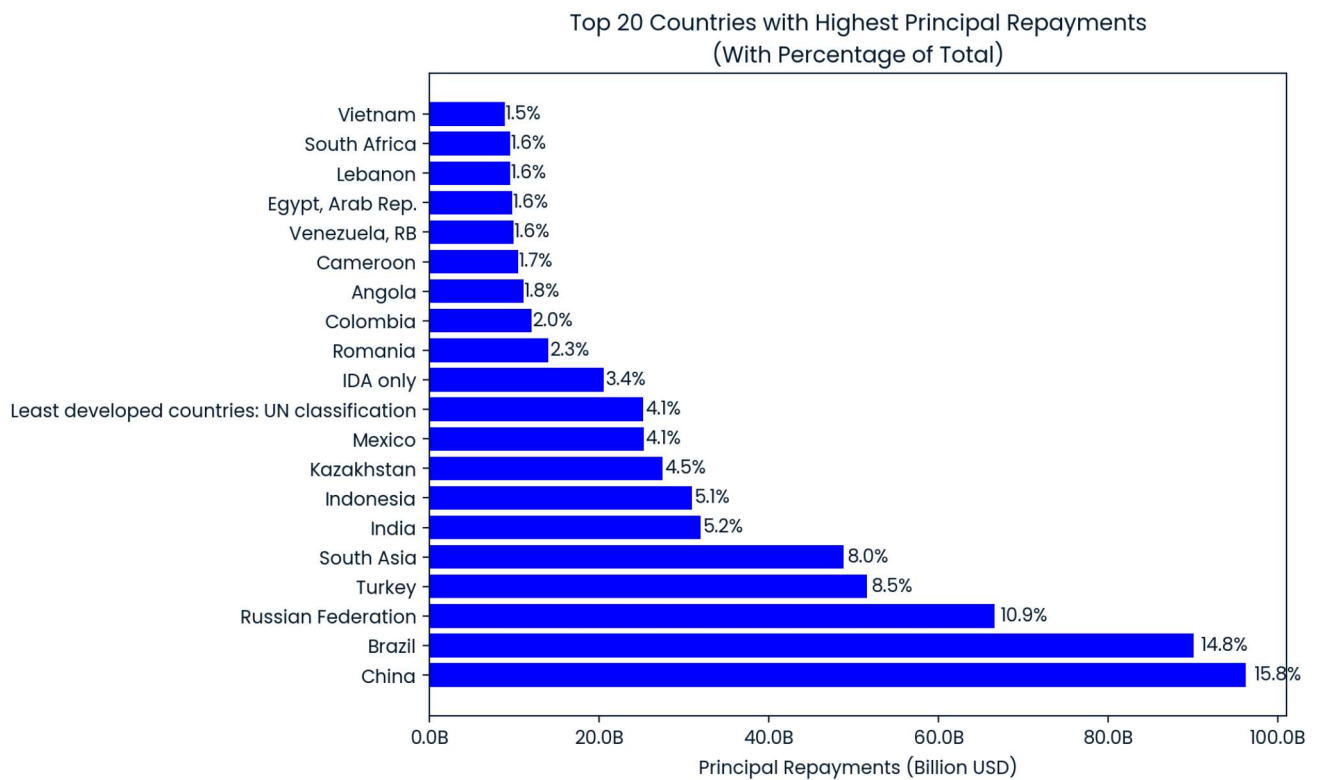
plt.figure(figsize=(10, 6))
plt.barh(highest_repayments_df['country_name'],
         highest_repayments_df['highest_repayments'],
         color='blue')

# Format x-axis values as billions
formatter = ticker.FuncFormatter(lambda x, _: f'{x/1e9:.1f}B')
plt.gca().xaxis.set_major_formatter(formatter)

# Add value labels (repayment percentage) on each bar
for i, (value, pct) in enumerate(zip(highest_repayments_df['highest_repayments'],
                                   highest_repayments_df['repayment_percentage'])):
    plt.text(value + 0.01 * value, i, f'{pct:.1f}%', va='center', fontsize=10)

plt.xlabel('Principal Repayments (Billion USD)')
plt.title('Top 20 Countries with Highest Principal Repayments\n(With Percentage of Total)')
plt.tight_layout()
plt.show()

```



```
-- select country_name, indicator_name, debt where indicator_code = 'DT.INT.DLXF.CD'
```

```
SELECT country_name, indicator_name, debt
FROM international_debt
WHERE indicator_code = 'DT.INT.DLXF.CD'
ORDER BY debt DESC
LIMIT 20;
```

...	↑↓	country_name	...	↑↓	indicator_name	...	↑↓	debt	...	↑↓	
	0	Mexico			Interest payments on external debt, long-ter...			19267966623.4			
	1	China			Interest payments on external debt, long-ter...			17866548651.4			
	2	Brazil			Interest payments on external debt, long-ter...			17001653109.2			
	3	South Asia			Interest payments on external debt, long-ter...			15931560388.1			
	4	Russian Federation			Interest payments on external debt, long-ter...			13929825315.6			
	5	Turkey			Interest payments on external debt, long-ter...			12095568642.2			
	6	India			Interest payments on external debt, long-ter...			11419475865.8			
	7	Indonesia			Interest payments on external debt, long-ter...			10190097793.8			
	8	IDA only			Interest payments on external debt, long-ter...			7269408633.8			
	9	Least developed countries: UN classification			Interest payments on external debt, long-ter...			7085064805.2			
	10	Kazakhstan			Interest payments on external debt, long-ter...			5333500397.9			
	11	South Africa			Interest payments on external debt, long-ter...			4872276053.4			
	12	Colombia			Interest payments on external debt, long-ter...			4564574602.1			
	13	Venezuela, RB			Interest payments on external debt, long-ter...			4159342365.6			
	14	Georgia			Interest payments on external debt, long-ter...			3087990080.3			
	15	Ukraine			Interest payments on external debt, long-ter...			3001506404.4			

Rows: 20


 Expand

```

plt.figure(figsize=(10, 6)) # Set chart size

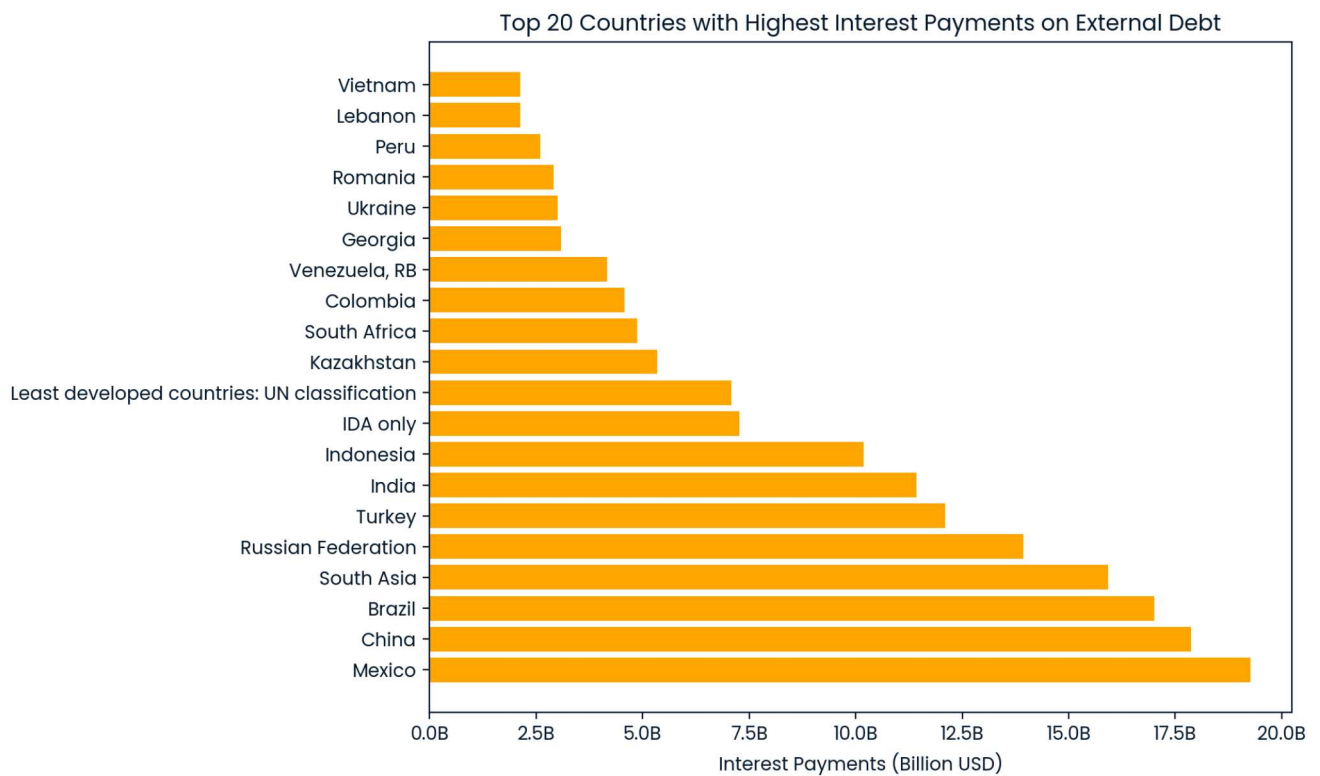
# Create horizontal bars for interest payments
plt.barh(highest_interest_df['country_name'],
         highest_interest_df['debt'],
         color='orange')

# Format x-axis values in billions (e.g., 25.3B)
formatter = ticker.FuncFormatter(lambda x, _: f'{x/1e9:.1f}B')
plt.gca().xaxis.set_major_formatter(formatter)

# Add labels and title
plt.xlabel('Interest Payments (Billion USD)')
plt.title('Top 20 Countries with Highest Interest Payments on External Debt')

plt.tight_layout()
plt.show()

```



```
-- countries with highest interest payments that also fall under highest principal repayments
-- Find top 20 countries by total principal repayments
```

```
WITH top_repayment_countries AS (
  SELECT country_name
  FROM international_debt
  WHERE indicator_code = 'DT.AMT.DLXF.CD'
  ORDER BY debt DESC
  LIMIT 20
)

-- Interest payments for those countries
SELECT d.country_name, d.indicator_name, d.debt AS interest_payments
FROM international_debt d
JOIN top_repayment_countries r ON d.country_name = r.country_name
WHERE d.indicator_code = 'DT.INT.DLXF.CD'
ORDER BY d.debt DESC;
```

...	↑↓	country_name	...	↑↓	indicator_name	...	↑↓	interest_pay...	...	↑↓
	0	Mexico			Interest payments on external debt, long-ter...			19267966623.4		
	1	China			Interest payments on external debt, long-ter...			17866548651.4		
	2	Brazil			Interest payments on external debt, long-ter...			17001653109.2		
	3	South Asia			Interest payments on external debt, long-ter...			15931560388.1		
	4	Russian Federation			Interest payments on external debt, long-ter...			13929825315.6		
	5	Turkey			Interest payments on external debt, long-ter...			12095568642.2		
	6	India			Interest payments on external debt, long-ter...			11419475865.8		
	7	Indonesia			Interest payments on external debt, long-ter...			10190097793.8		
	8	IDA only			Interest payments on external debt, long-ter...			7269408633.8		
	9	Least developed countries: UN classification			Interest payments on external debt, long-ter...			7085064805.2		
	10	Kazakhstan			Interest payments on external debt, long-ter...			5333500397.9		
	11	South Africa			Interest payments on external debt, long-ter...			4872276053.4		
	12	Colombia			Interest payments on external debt, long-ter...			4564574602.1		
	13	Venezuela, RB			Interest payments on external debt, long-ter...			4159342365.6		
	14	Romania			Interest payments on external debt, long-ter...			2906875294.6		
	15	Lebanon			Interest payments on external debt, long-ter...			2131942998.7		

Rows: 20

Expand

```

# Convert interest payments to billions for readability
highest_interest_in_top_repayments_df['interest_billions'] = highest_interest_in_top_repayments_df['interest_payments'] / 1e9

# Sort by interest payments descending
df_sorted = highest_interest_in_top_repayments_df.sort_values(by='interest_billions', ascending=True)

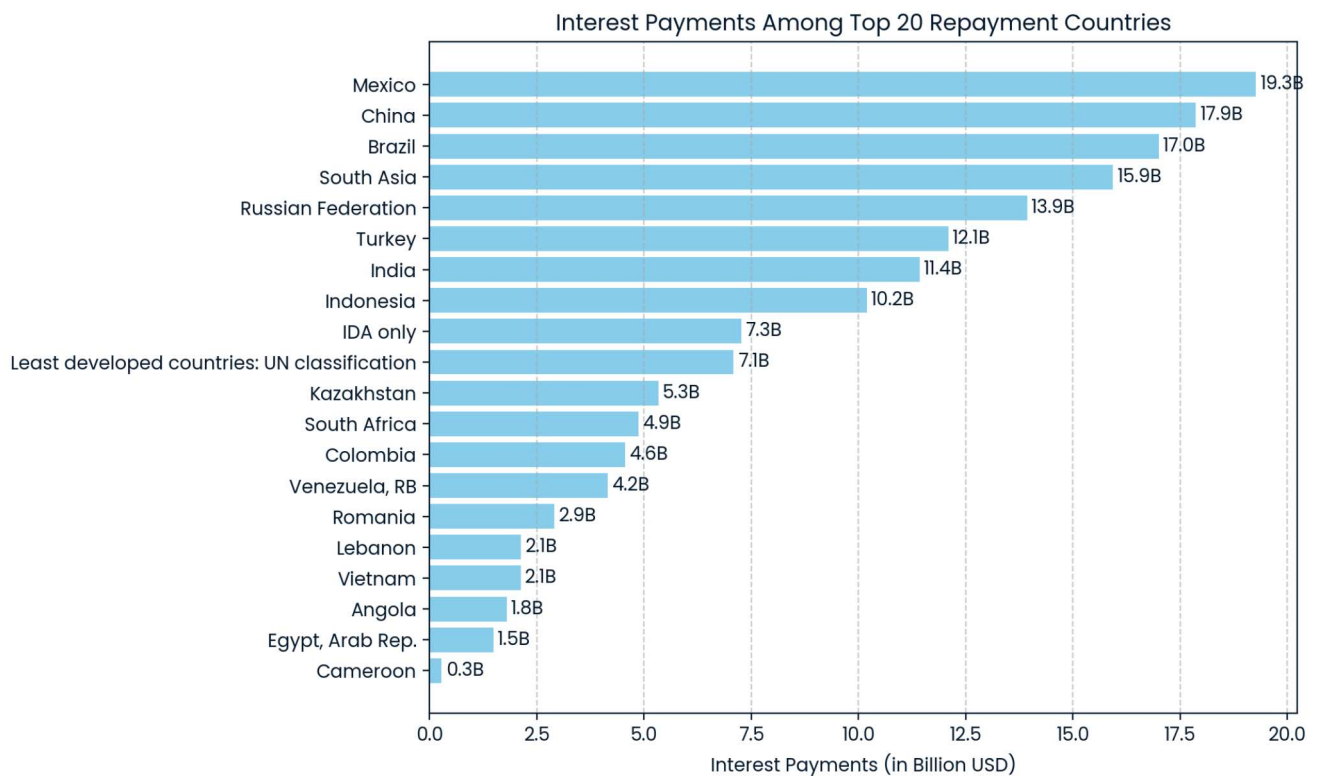
# Plot
plt.figure(figsize=(10, 6))
bars = plt.barh(df_sorted['country_name'], df_sorted['interest_billions'], color='skyblue')

# Add text labels
for bar in bars:
    plt.text(bar.get_width() + 0.1, bar.get_y() + bar.get_height()/2,
             f"{bar.get_width():.1f}B", va='center')

# Titles and Labels
plt.xlabel('Interest Payments (in Billion USD)')
plt.title('Interest Payments Among Top 20 Repayment Countries')
plt.tight_layout()
plt.grid(axis='x', linestyle='--', alpha=0.6)

plt.show()

```



```
--Breakdown of disbursements for 5 countries with the highest and lowest disbursements
```

```
(
  SELECT country_name,
    SUM(CASE
      WHEN indicator_code = 'DT.DIS.DLXF.CD' THEN debt
      ELSE 0
    END) AS total_disbursements,
    SUM(CASE
      WHEN indicator_code = 'DT.AMT.DLXF.CD' THEN debt
      ELSE 0
    END) AS total_repayments

  FROM international_debt
  GROUP BY country_name
  ORDER BY total_disbursements DESC
  LIMIT 5
)

UNION ALL

(
  SELECT country_name,
    SUM(CASE
      WHEN indicator_code = 'DT.DIS.DLXF.CD' THEN debt
      ELSE 0
    END) AS total_disbursements,
    SUM(CASE
      WHEN indicator_code = 'DT.AMT.DLXF.CD' THEN debt
      ELSE 0
    END) AS total_repayments

  FROM international_debt
  GROUP BY country_name
  ORDER BY total_disbursements ASC
  LIMIT 5
);
```

...	↑↓	country_name	...	↑↓	total_disburseme...	...	↑↓	total_repay...	...	↑↓
	0	Least developed countries: UN classification			40160766261.6			25197029299.4		
	1	IDA only			34531188113.2			20483289208		
	2	South Asia			29306216064.7			48756295898.2		
	3	Cameroon			18186662060.4			10404814960.2		
	4	China			15692563746.1			96218620835.7		
	5	Turkmenistan			0			29132060.9		
	6	Somalia			2456971.6			32985034.3		
	7	Tonga			5529604.8			10369910.6		
	8	Eritrea			7203213.6			31110077.8		
	9	Solomon Islands			8645249.9			30749703.9		

Rows: 10


Expand

```

SELECT
  country_name,
  indicator_name,
  SUM(debt) AS total_disbursement
FROM international_debt
WHERE country_name IN (
  'Least developed countries: UN classification',
  'IDA only',
  'South Asia',
  'Cameroon',
  'China',
  'Turkmenistan',
  'Somalia',
  'Tonga',
  'Eritrea',
  'Solomon Islands'
)
AND indicator_code LIKE 'DT.DIS%'           -- Keep only disbursements
AND indicator_code != 'DT.DIS.DLXF.CD'     -- Exclude the total disbursement row
GROUP BY country_name, indicator_name
ORDER BY country_name, total_disbursement DESC;

```

...	↑↓	country_name	...	↑↓	indicator_name	...	↑↓	total_disburse...	...	↑↓
	0	Cameroon			PPG, official creditors (DIS, current US\$)			17940581035.6		
	1	Cameroon			PPG, bilateral (DIS, current US\$)			17090544247.5		
	2	Cameroon			PPG, multilateral (DIS, current US\$)			850036788.1		
	3	Cameroon			PPG, private creditors (DIS, current US\$)			246081024.8		
	4	Cameroon			PPG, commercial banks (DIS, current US\$)			196167124.8		
	5	Cameroon			PPG, other private creditors (DIS, current US\$)			49913900		
	6	China			PPG, private creditors (DIS, current US\$)			4111062474		
	7	China			PPG, commercial banks (DIS, current US\$)			3777050273.3		
	8	China			PPG, multilateral (DIS, current US\$)			3079501272.1		
	9	China			PPG, official creditors (DIS, current US\$)			3079501272.1		
	10	China			PPG, other private creditors (DIS, current US\$)			334012200.7		
	11	Eritrea			PPG, official creditors (DIS, current US\$)			7203213.6		
	12	Eritrea			PPG, multilateral (DIS, current US\$)			4524411		
	13	Eritrea			PPG, bilateral (DIS, current US\$)			2678802.6		
	14	IDA only			PPG, official creditors (DIS, current US\$)			33258230166.9		
	15	IDA only			PPG, bilateral (DIS, current US\$)			16882903064.4		

Rows: 37

Expand

```

import pandas as pd
import matplotlib.pyplot as plt

def plot_disbursement_pie_charts(disbursement_breakdown_df, threshold_percent=5):
    pie_colors = ['blue', 'orange', 'purple', 'lightblue', 'yellow', 'magenta', 'gray']

    countries = disbursement_breakdown_df['country_name'].unique()

    for country in countries:
        data = disbursement_breakdown_df[disbursement_breakdown_df['country_name'] == country].copy()
        total = data['total_disbursement'].sum()
        data['percent'] = 100 * data['total_disbursement'] / total

        # Split into large and small contributions
        large = data[data['percent'] >= threshold_percent]
        small = data[data['percent'] < threshold_percent]

        # Combine small into 'Other'
        if not small.empty:
            other_row = pd.DataFrame({
                'country_name': [country],
                'indicator_name': ['Other'],
                'total_disbursement': [small['total_disbursement'].sum()],
                'percent': [small['percent'].sum()]
            })
            data = pd.concat([large, other_row], ignore_index=True)

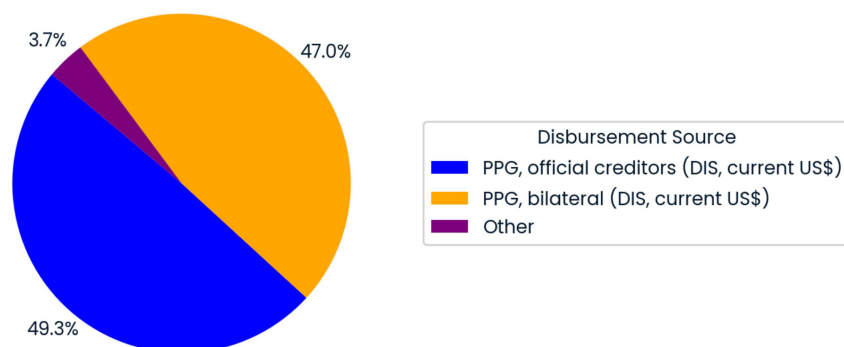
        # Sort by disbursement
        data = data.sort_values(by='total_disbursement', ascending=False)

        # Plot pie chart
        plt.figure(figsize=(7, 7))
        plt.pie(
            data['total_disbursement'],
            labels=None,
            autopct='%1.1f%%',
            pctdistance=1.15,
            colors=pie_colors[:len(data)],
            startangle=140,
            textprops={'fontsize': 10}
        )
        plt.legend(data['indicator_name'], title="Disbursement Source", bbox_to_anchor=(1.05, 0.5), loc='center left')
        plt.title(f'Disbursement Breakdown for {country}')
        plt.tight_layout()
        plt.show()

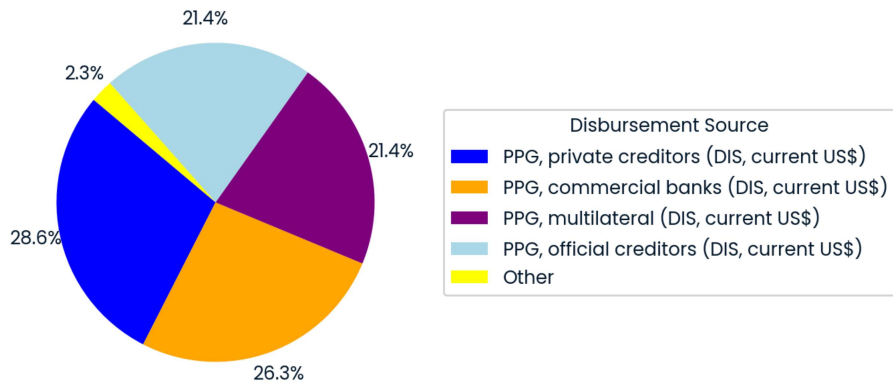
plot_disbursement_pie_charts(disbursement_breakdown_df)

```

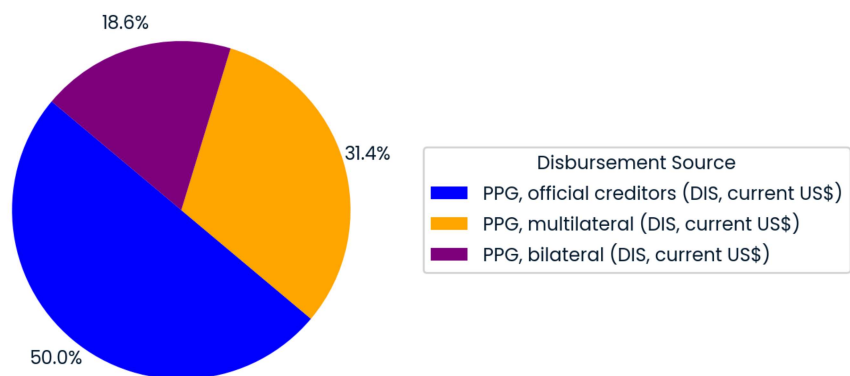
Disbursement Breakdown for Cameroon



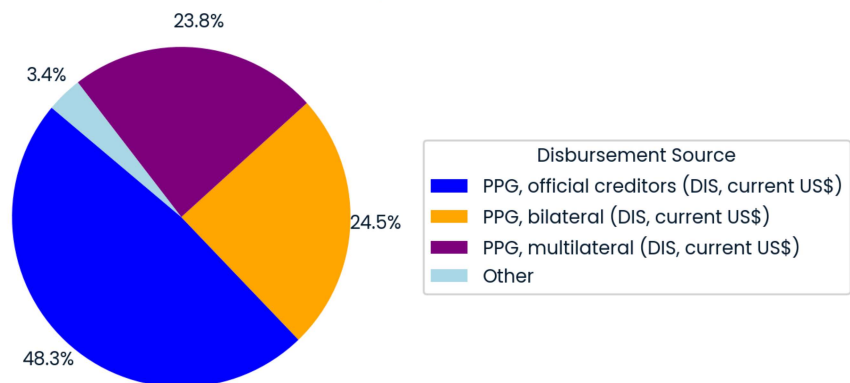
Disbursement Breakdown for China



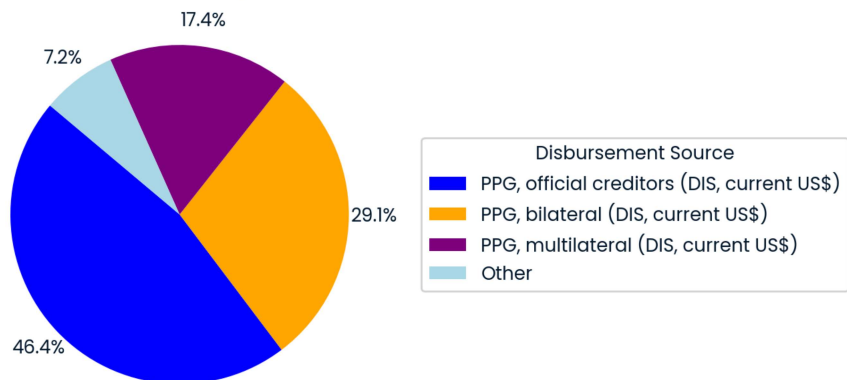
Disbursement Breakdown for Eritrea



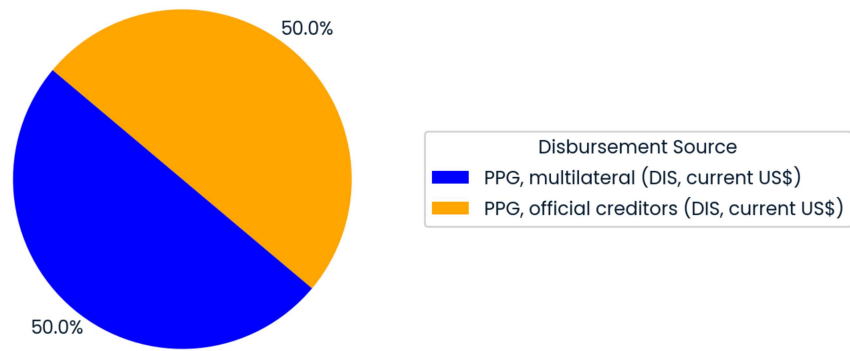
Disbursement Breakdown for IDA only



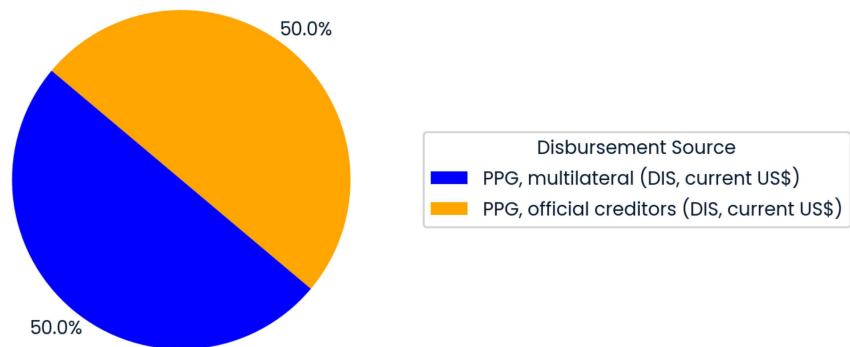
Disbursement Breakdown for Least developed countries: UN classification



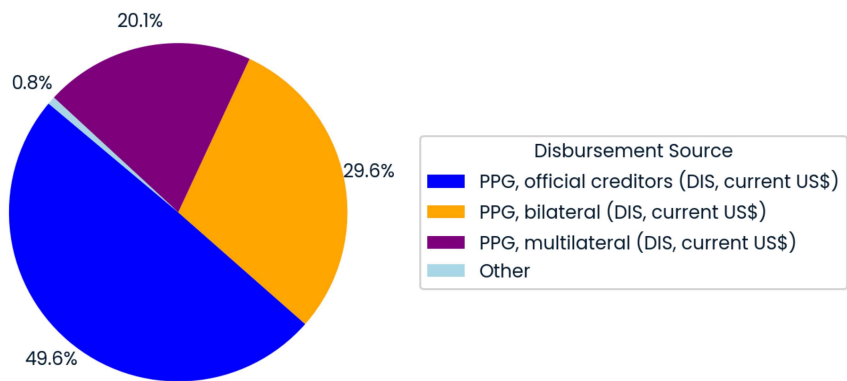
Disbursement Breakdown for Solomon Islands



Disbursement Breakdown for Somalia



Disbursement Breakdown for South Asia



Disbursement Breakdown for Tonga

